

Baixiang Huang

[✉ Email](#) | [🌐 Website](#) | [🎓 Google Scholar](#) | [🐙 GitHub](#) | [🌐 LinkedIn](#)

Research interests: Alignment • Agentic RL for Tool Use • Evals • Continual Learning • Personalization

EDUCATION

Emory University

Jan 2023 – Present

Ph.D. in Computer Science

Atlanta, GA

Advisor: [Kai Shu](#)

National University of Singapore

Aug 2020 – Feb 2022

Master's Degree in Computer Science

Singapore

Advisor: [Bryan Hooi](#)

PROFESSIONAL EXPERIENCE

Google

Sep 2026 – Present

Student Researcher — Semantic Cache Poisoning in LLM Serving

San Jose, CA

- Investigating semantic cache poisoning attacks in LLM serving systems, focusing on how adversarial cache entries can manipulate retrieved responses and downstream generations.
- Targeting Google Search AI Overviews, where cached content can influence generated answers.

Amazon

May 2026 – Aug 2026

Applied Scientist Intern — SOP Alignment for Tool-Use Agents

Santa Clara, CA

- Introduced Procedural Editing and developed a closed-loop post-training framework that repairs Standard Operating Procedure (SOP) violations in tool-use LLM agents.
- Localized the first divergent decision, optimized corrections at the agent's recorded decision state, and reran trajectories after each update to identify and repair downstream failures.
- Evaluated the proposed method across different models and ReAct and function-calling scaffolds, achieving the highest success rate in all four settings while preserving general capabilities.

Amazon

May 2025 – Aug 2025

Applied Scientist Intern — LLM Personalization [PDF] [GitHub] [Project Website]

Palo Alto, CA

- Introduced Personalization Editing, a editing-based method that encodes clustered user preferences into persistent model behavior while preserving unrelated capabilities.
- Designed an in-situ evaluation suite for explicit, implicit, and multi-turn preference following, and developed the associated training and evaluation pipeline.
- Outperformed fine-tuning on personalization accuracy and efficiency and prompting baselines on robustness while retaining general capabilities.

PUBLICATIONS AND PREPRINTS

Peer-Reviewed Publications

[1] **Agents' Last Exam**, *NeurIPS 2026 Datasets & Benchmarks Track*

Yiyu Sun, and many others including **Baixiang Huang**.

[PDF] [GitHub] [Project Website]

[2] **DrugReason: Dynamic Multi-View Reasoning over Knowledge Graph and Language Evidence for Drug Repurposing**, *EMNLP 2026*

Zijie Liu, Hongxuan Li, Zhen Tan, Jinhao Duan, **Baixiang Huang**, Zunpeng Liu, Kai Shu, Tianlong Chen. [PDF]

[3] **Model Editing as a Double-Edged Sword: Steering Agent Ethical Behavior Toward Beneficence or Harm**, *AAAI 2026 Oral*

Baixiang Huang, Zhen Tan, Haoran Wang, Zijie Liu, Dawei Li, Ali Payani, Huan Liu, Tianlong Chen, Kai Shu.

[PDF] [GitHub] [Project Website]

- [4] **Can Editing LLMs Inject Harm?**, *AAAI 2026*
Canyu Chen*, **Baixiang Huang***, Zekun Li, Zhaorun Chen, Shiyang Lai, Xiong Xiao Xu, Jia-Chen Gu, Jindong Gu, Huaxiu Yao, Chaowei Xiao, Xifeng Yan, William Wang, Philip Torr, Dawn Song, Kai Shu.
(* denotes equal contribution)
[\[PDF\]](#) [\[GitHub\]](#) [\[Project Website\]](#)
- [5] **A Survey of Agent Memory in the Second Half: Towards Self-Evolving and Long-Horizon Agents**, *TMLR 2026*
Wei-Chieh Huang, and many others including **Baixiang Huang**. [\[PDF\]](#)
- [6] **Privacy-Aware Decoding: Mitigating Privacy Leakage of Large Language Models in Retrieval-Augmented Generation**, *KDD 2026*
Haoran Wang, Xiong Xiao Xu, **Baixiang Huang**, Kai Shu.
[\[PDF\]](#) [\[GitHub\]](#)
- [7] **Can Knowledge Editing Really Correct Hallucinations?**, *ICLR 2025*
Baixiang Huang, Canyu Chen, Xiong Xiao Xu, Ali Payani, Kai Shu.
[\[PDF\]](#) [\[GitHub\]](#) [\[Project Website\]](#)
- [8] **SST: Multi-Scale Hybrid Mamba-Transformer Experts for Long-Short Range Time Series Forecasting**, *CIKM 2025*
Xiong Xiao Xu, Canyu Chen, Yueqing Liang, **Baixiang Huang**, Guangji Bai, Liang Zhao, Kai Shu.
[\[PDF\]](#) [\[GitHub\]](#)
- [9] **Can Large Language Models Identify Authorship?**, *EMNLP Findings 2024*
Baixiang Huang, Canyu Chen, Kai Shu.
[\[PDF\]](#) [\[GitHub\]](#) [\[Project Website\]](#)
- [10] **Authorship Attribution in the Era of LLMs: Problems, Methodologies, and Challenges**, *ACM SIGKDD Explorations 2024*
Baixiang Huang, Canyu Chen, Kai Shu.
[\[PDF\]](#) [\[GitHub\]](#) [\[Project Website\]](#)
- [11] **TAP: A Comprehensive Data Repository for Traffic Accident Prediction in Road Networks**, *ACM SIGSPATIAL 2023*
Baixiang Huang, Bryan Hooi, Kai Shu.
[\[PDF\]](#) [\[GitHub\]](#)

Preprints

- [1] **Rethinking the Value of Multi-Agent Workflow: A Strong Single Agent Baseline**, *arXiv preprint, 2026*
Jiawei Xu, Arief Koesdwiady, Sisong Bei, Yan Han, **Baixiang Huang**, Dakuo Wang, Yutong Chen, Zheshen Wang, Peihao Wang, Pan Li, Ying Ding. [\[PDF\]](#)
- [2] **Towards Effective Model Editing for LLM Personalization**, *arXiv preprint, 2025*
Baixiang Huang, Limeng Cui, Jiapeng Liu, Haoran Wang, Jiawei Xu, Zhu Yue Tan, Yutong Chen, Chen Luo, Yi Liu, Kai Shu. [\[PDF\]](#) [\[GitHub\]](#) [\[Project Website\]](#)
- [3] **Who's Your Judge? On the Detectability of LLM-Generated Judgments**, *arXiv preprint, 2025*
Dawei Li, Zhen Tan, Chengshuai Zhao, Bohan Jiang, **Baixiang Huang**, Pingchuan Ma, Abdullah Alnaibari, Kai Shu, Huan Liu. [\[PDF\]](#) [\[GitHub\]](#) [\[Project Website\]](#)

TECHNICAL SKILLS

- **Research & Methods:** Agent post-training, agentic RL, alignment and evals, tool use, continual learning, model editing, personalization
- **ML Systems:** PyTorch, TensorFlow, vLLM; LLM training, inference, and evaluation pipelines
- **Programming:** Python, C++, Java